

EC3108 FINANCIAL ECONOMIC DYNAMICS

Foundations: Dynamical Systems Toolkit

Lecture Notes

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This is a standalone lecture notes covering the Foundations material, and it builds the technical apparatus the course draws on: difference equations in discrete time, phase diagrams and saddle paths in continuous time, the Hamiltonian and its costate variable, and the elementary stochastic processes used from Week 2 onward. Where the method of an earlier lecture is needed for comparison - the discrete-time Euler-equation method, in particular - it is briefly restated here rather than assumed, so that this lecture can be read on its own. Every proof is broken into small, separately justified steps, and every substantial result is followed by a plain-English restatement.

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How to read these notes

These notes use seven kinds of box, consistently, throughout every lecture in this sequence. Knowing what each one is for before you start will save you time.

Quick refresher

A **Quick refresher**. Background mathematics that you have almost certainly seen in a previous course but might not have used recently. If the refresher is familiar, skip it. If it is not, read it before the section that follows, because everything after it assumes it.

In plain English

An **In plain English** box. Every substantial equation or theorem in these notes is followed by one of these: the same content, restated as a sentence you could say out loud to a friend who has not seen the algebra.

On method

An **On method** box. This course solves every dynamic optimisation problem one of two ways — an Euler equation in discrete time, or a Hamiltonian and its costate variable in continuous time — and never uses a value function or the Bellman equation. These boxes flag exactly what the method is doing at each step, and how the discrete- and continuous-time versions mirror each other.

Common confusion

A **Common confusion**. These flag the specific place where students most often go wrong — usually because two ideas look similar but are not the same idea.

Static recap

A **Static recap**. The one-period result you already know, stated precisely so that the notation is fixed before we start changing it.

The assumption being bought

The assumption being bought: what the world must be like for a result's tractability to be earned honestly.

Where this reappears

A **Where this reappears** box. A pointer to the later week or unit that uses the result just derived. These are worth reading even on a first pass: they tell you why you are being asked to learn this now.

One more habit worth adopting: every proof in these notes is broken into numbered steps, and every step has a short parenthetical reason attached to it. When you get stuck on a proof, the question to ask is never “what is the next line of algebra” but “what is the *reason* for the next line”.

1 Difference equations

We begin in discrete time, which is where most of the finance in this course lives.

1.1 First-order linear difference equations

Definition 1.1. A first-order linear difference equation with constant coefficients is

$$x_{t+1} = ax_t + b, \quad a, b \in \mathbb{R}, \quad x_0 \text{ given.} \quad (1)$$

Proposition 1.2 (Solution and steady state). *If $a \neq 1$, equation (1) has the unique steady state $x^* = b/(1 - a)$ and general solution*

$$x_t = a^t (x_0 - x^*) + x^*. \quad (2)$$

If $a = 1$, there is no steady state (unless $b = 0$) and $x_t = x_0 + bt$.

Proof. Step 1: find the steady state. It solves $x^* = ax^* + b$, giving $x^* = b/(1 - a)$ (requires $a \neq 1$ to divide).

Step 2: work in deviations from the steady state. Define $y_t \equiv x_t - x^*$. Substituting into (1),

$$y_{t+1} + x^* = a(y_t + x^*) + b \implies y_{t+1} = ay_t + [ax^* + b - x^*] = ay_t,$$

the bracket vanishing by the definition of x^* (this is the whole point of working in deviations: the additive constant b disappears entirely).

Step 3: iterate the resulting homogeneous equation. $y_{t+1} = ay_t$ gives $y_t = a^t y_0$ directly, which is (2). $\square$

In plain English

The trick in this proof — subtract off the steady state and look at what is left — is one you will use constantly for the rest of the course. It works because a linear equation's deviation from its own fixed point obeys a *simpler* equation (no additive constant) than the original.

The behaviour of (2) is governed entirely by a :

Range of a	Behaviour	Convergence
$0 < a < 1$	monotone approach	yes
$-1 < a < 0$	damped oscillation	yes
$a = 1$	linear drift	no
$a = -1$	two-cycle, constant amplitude	no
$a > 1$	monotone divergence	no
$a < -1$	explosive oscillation	no

In plain English

a answers two separate questions at once. Its *sign* says whether the path overshoots the steady state each period (negative a : the deviation flips sign every period, so the path oscillates around x^*) or approaches it from one side (positive a). Its *magnitude* says whether the deviation is shrinking ($|a| < 1$) or growing ($|a| > 1$) each period. Four combinations of

sign and magnitude, four qualitatively different pictures.

Corollary 1.3 (Stability condition). *The steady state of (1) is globally asymptotically stable if and only if $|a| < 1$.*

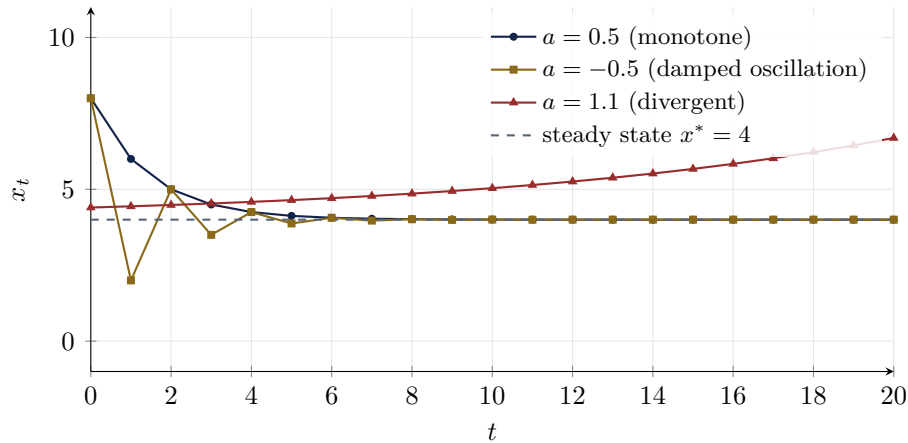


Figure 1: Sample paths of $x_{t+1} = ax_t + b$ with $x^* = 4$. The sign of a decides whether the approach is monotone or oscillatory; $|a|$ decides whether there is an approach at all.

1.2 Second-order equations

Definition 1.4. A second-order linear difference equation is

$$x_{t+2} + a_1x_{t+1} + a_0x_t = c. \quad (3)$$

Quick refresher

Why guess $x_t = \lambda^t$? For a linear, constant-coefficient equation, if λ^t solves the homogeneous equation for some λ , then so does any constant multiple of it, and sums of solutions are also solutions (this is what “linear” buys you). The guess turns a *difference equation* in t into an *algebraic equation* in λ — the characteristic equation below — which is far easier to solve. This is exactly the same trick used for linear differential equations, and for exactly the same reason.

Guessing $x_t = \lambda^t$ for the homogeneous part gives the *characteristic equation*

$$\lambda^2 + a_1\lambda + a_0 = 0, \quad \lambda_{1,2} = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_0}}{2}. \quad (4)$$

Three cases arise.

1. **Distinct real roots** ($a_1^2 > 4a_0$): $x_t = A\lambda_1^t + B\lambda_2^t + x^*$.
2. **Repeated root** ($a_1^2 = 4a_0$), $\lambda = -a_1/2$: $x_t = (A + Bt)\lambda^t + x^*$.
3. **Complex conjugate roots** ($a_1^2 < 4a_0$), so that $\lambda = re^{\pm i\theta}$ with $r = \sqrt{a_0}$ and $\theta = \arccos(-a_1/(2\sqrt{a_0}))$:

$$x_t = r^t(A \cos \theta t + B \sin \theta t) + x^*.$$

The path *oscillates* with period $2\pi/\theta$, and the amplitude grows or decays according to whether $r \gtrless 1$.

In plain English

Complex roots sound alarming but are not: whenever they appear, always in a conjugate pair $re^{\pm i\theta}$, the imaginary parts cancel in the final formula and x_t is guaranteed real. All that survives from the complex arithmetic is r (how fast the oscillation's amplitude grows or shrinks) and θ (how fast it oscillates). You never need to carry an actual complex number through to the final answer.

Proposition 1.5 (Stability). *The steady state of (3) is asymptotically stable if and only if both roots lie strictly inside the unit circle, which holds if and only if*

$$1 + a_1 + a_0 > 0, \quad 1 - a_1 + a_0 > 0, \quad a_0 < 1.$$

Remark 1.6. Case 3 is the one that matters economically. Complex roots are how a linear model produces cycles *without* any cyclical forcing: the oscillation is a property of the system, not of what is done to it. This is a first, mild instance of the idea that dominates Unit 5 — that fluctuations can be generated internally rather than imposed from outside.

2 Present value as a difference equation

Now a genuinely financial application, and the first appearance of a result we shall use repeatedly.

2.1 The pricing equation

The no-arbitrage condition for an asset paying dividend D_{t+1} and requiring expected return r is

$$P_t = \frac{\mathbb{E}_t[P_{t+1} + D_{t+1}]}{1 + r} \iff \mathbb{E}_t[P_{t+1}] = (1 + r)P_t - \mathbb{E}_t[D_{t+1}]. \quad (5)$$

In plain English

The first form is the ordinary statement of no arbitrage: today's price is the discounted expected value of tomorrow's price plus tomorrow's dividend. The second form is exactly the same equation, rearranged to look like the difference equation (1) you have just studied, with the price playing the role of x_t .

Written the second way, this is a first-order linear difference equation in the price, with coefficient $a = 1 + r > 1$. By the corollary above, that root lies *outside* the unit circle: the homogeneous solution is explosive.

2.2 Fundamental and bubble solutions

The general solution of (5) is the sum of a particular solution and the general solution of the homogeneous equation $\mathbb{E}_t[B_{t+1}] = (1 + r)B_t$:

$$P_t = \underbrace{\sum_{j=1}^{\infty} \frac{\mathbb{E}_t[D_{t+j}]}{(1+r)^j}}_{\text{fundamental } F_t} + \underbrace{B_t}_{\text{bubble}}, \quad \mathbb{E}_t[B_{t+j}] = (1+r)^j B_t. \quad (6)$$

In plain English

The fundamental piece F_t is the textbook present-value formula: discount every future dividend back to today and add them up. The bubble piece B_t is any process that *itself* grows at exactly rate r in expectation — so that adding it to F_t still satisfies the no-arbitrage equation (5), since F_t alone already satisfies it and the equation is linear. Nothing about no-arbitrage, by itself, says B_t must be zero.

Example 2.1 (Constant dividend with a bubble). Let $r = 0.05$ and $D_t = 2$ for all t . The fundamental value is $F = D/r = 2/0.05 = 40$. Now suppose $B_0 = 1$. Then $B_t = 1.05^t$, so

$$B_{10} = 1.629, \quad B_{20} = 2.653, \quad B_{50} = 11.467.$$

The price is 41 at date 0 and 51.47 at date 50, and the bubble component grows without bound as a proportion of the price. Note that nothing in (5) is violated at any date: the bubble is fully consistent with no-arbitrage and rational expectations at every point on the path.

2.3 The transversality condition

To recover the familiar textbook formula $P_t = F_t$, one must impose

$$\lim_{T \rightarrow \infty} \frac{\mathbb{E}_t[P_{t+T}]}{(1+r)^T} = 0, \quad (7)$$

which rules out $B_t \neq 0$ by fiat.

In plain English

(7) says: the price, discounted far enough into the future, must shrink to nothing. A bubble violates this — by construction it grows at rate r , exactly cancelling the discounting, so its discounted value never shrinks. Imposing (7) is therefore mathematically equivalent to simply declaring $B_t = 0$.

The assumption being bought

The present-value formula you were taught statically is the *particular* solution of a difference equation whose homogeneous solution was discarded by assumption. The static treatment cannot even state the alternative, because the alternative is a *path*, not a number. This is the cleanest illustration in the course of what dynamics adds: an entire family of solutions invisible from inside the static model.

Where this reappears

The bubble solution is taken up properly in Unit 3 (Week 7), alongside the Campbell–Shiller log-linear identity. The observation that $(1+r)$ is an unstable root, and that the economically sensible path is therefore the one selected by a terminal condition rather than an initial condition, is exactly the saddle-path logic of the next section — and, in continuous time with an optimising agent rather than a pricing identity, it is exactly the transversality condition of the Hamiltonian method in §4.

3 Continuous time, phase diagrams and saddle paths

3.1 Scalar autonomous equations

Definition 3.1. An autonomous first-order differential equation is $\dot{x} = f(x)$. A steady state satisfies $f(x^*) = 0$.

Proposition 3.2 (Local stability). *If f is continuously differentiable, the steady state x^* is locally asymptotically stable if $f'(x^*) < 0$ and unstable if $f'(x^*) > 0$.*

Proof. Linearise: writing $y = x - x^*$, a first-order Taylor expansion gives $\dot{y} \approx f'(x^*)y$, whose solution is $y(t) = y(0)e^{f'(x^*)t}$. This decays to zero if and only if $f'(x^*) < 0$. $\square$

In plain English

Near the steady state, any smooth f looks approximately like a straight line through x^* with slope $f'(x^*)$ — that is what “linearise” means. A negative slope means f is pulling x back toward x^* whenever it strays; a positive slope means f is pushing x further away. Everything about local stability is contained in the sign of one number.

3.2 Planar linear systems

Most economic applications involve two variables. Consider

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}. \quad (8)$$

The eigenvalues of $\mathbf{A}$ solve $\lambda^2 - \tau\lambda + \Delta = 0$, where $\tau = \text{tr } \mathbf{A}$ and $\Delta = \det \mathbf{A}$. Everything about the local behaviour follows from (τ, Δ) :

Condition	Type	Behaviour
$\Delta < 0$	saddle point	one stable, one unstable direction
$\Delta > 0, \tau < 0, \tau^2 > 4\Delta$	stable node	monotone convergence
$\Delta > 0, \tau < 0, \tau^2 < 4\Delta$	stable focus	spiral convergence
$\Delta > 0, \tau > 0, \tau^2 > 4\Delta$	unstable node	monotone divergence
$\Delta > 0, \tau > 0, \tau^2 < 4\Delta$	unstable focus	spiral divergence
$\Delta > 0, \tau = 0$	centre	closed orbits

Quick refresher

Why the trace and determinant are enough. The eigenvalues of a 2×2 matrix solve $\lambda^2 - \tau\lambda + \Delta = 0$ regardless of what the individual entries $a_{11}, \dots, a_{22}$ are — only their combination into τ (the sum of the diagonal) and Δ (the determinant) matters for stability. A negative Δ means the two eigenvalues have opposite sign (one positive root, one negative, since their product is Δ) — automatically a saddle, regardless of the trace. A positive Δ with negative τ means both eigenvalues are negative (their sum is τ , their product Δ , both conditions force both roots negative) — stable. The sign of the discriminant $\tau^2 - 4\Delta$ then decides real versus complex roots, i.e. monotone versus oscillatory.

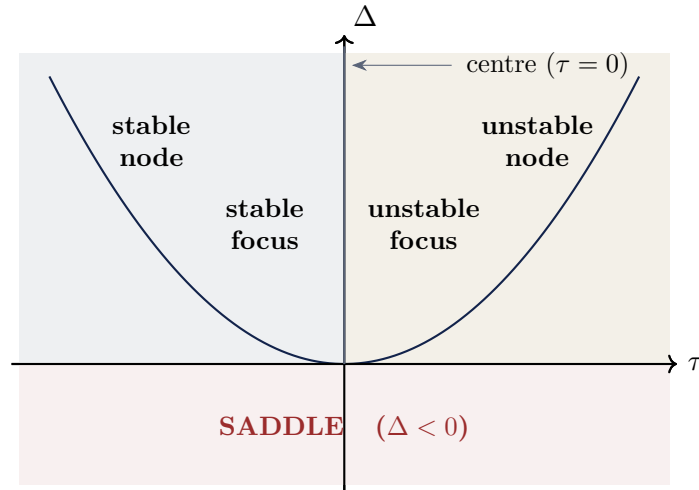


Figure 2: The trace–determinant plane. The parabola $\tau^2 = 4\Delta$ separates monotone from oscillatory behaviour; the horizontal axis separates stable from unstable; the entire half-plane $\Delta < 0$ is saddle territory. This is the single diagram against which every Hamiltonian–costate system in this course will be checked.

3.3 A financial saddle point

The saddle case looks pathological but is the workhorse of asset-price dynamics. Consider an asset whose dividend mean-reverts and whose price satisfies a no-arbitrage condition in continuous time:

$$\dot{P} = rP - D, \quad \dot{D} = g(\bar{D} - D), \quad r, g > 0. \quad (9)$$

The first equation is the continuous-time analogue of (5): the required return r must be delivered by the dividend yield plus the capital gain, so $\dot{P}/P + D/P = r$.

The steady state is $D^* = \bar{D}$, $P^* = \bar{D}/r$. The Jacobian is

$$\mathbf{J} = \begin{pmatrix} r & -1 \\ 0 & -g \end{pmatrix}, \quad \Delta = \det \mathbf{J} = -rg < 0,$$

so the system is a saddle, with eigenvalues $\lambda_1 = r > 0$ and $\lambda_2 = -g < 0$.

Proposition 3.3 (The saddle path is the present-value formula). *The stable manifold of (9) is the line*

$$P = \frac{\bar{D}}{r} + \frac{D - \bar{D}}{r + g}. \quad (10)$$

Proof. The eigenvector associated with $\lambda_2 = -g$ solves $(\mathbf{J} + g\mathbf{I})\mathbf{v} = 0$:

$$\begin{pmatrix} r + g & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_P \\ v_D \end{pmatrix} = 0 \implies v_D = (r + g)v_P.$$

The stable manifold is the line through the steady state in the direction $\mathbf{v} = (1, r + g)'$, that is $(D - \bar{D}) = (r + g)(P - \bar{D}/r)$, which rearranges to (10). $\square$

Remark 3.4 (Why this matters). Compute the present value directly. Since $\mathbb{E}_t[D_{t+s}] = \bar{D} + (D_t - \bar{D})e^{-gs}$,

$$P_t = \int_0^\infty e^{-rs} \mathbb{E}_t[D_{t+s}] ds = \frac{\bar{D}}{r} + \frac{D_t - \bar{D}}{r + g},$$

which is exactly (10). **The saddle path is the present-value formula.** The two derivations — one geometric, from the stable eigenvector, and one by integration — give the same object. Students who see this once tend not to forget what a saddle path is.

Example 3.5 (Numbers). With $\bar{D} = 2$, $r = 0.05$ and $g = 0.30$, the steady-state price is $P^* = 40$. If the dividend is currently $D = 2.5$, the saddle path gives

$$P = 40 + \frac{0.5}{0.35} = 41.43.$$

The price is above its steady state, but only by 1.43 rather than the 10 that capitalising the current dividend at r would suggest — because the dividend is expected to revert.

3.4 Jump variables and the selection of the path

The saddle has one unstable direction. If the system starts at a point that does not lie exactly on the stable manifold, it diverges: either the price explodes, or it falls towards zero and beyond. Neither is economically admissible.

Quick refresher

A thermostat, as an analogy for a jump variable. A room's temperature cannot change instantly — it drifts slowly, governed by however much heat is being added relative to how much escapes. A thermostat setting, by contrast, *can* change instantly: you turn the dial, and the target changes right away, even though the room itself has not caught up yet. The dividend in (9) is like the room temperature (predetermined, moves continuously); the price is like the thermostat dial (free to jump). The saddle-path logic says: the dial gets set, instantly, to whichever value puts the whole system on a path that does not run away to infinity or below zero.

The resolution is that P is a *jump variable*. Unlike the dividend — which is predetermined, inherited from the past, and cannot move discontinuously — the price is free to move instantly to whatever value the market assigns it. The economy therefore selects the value of P that places the system exactly on the stable manifold.

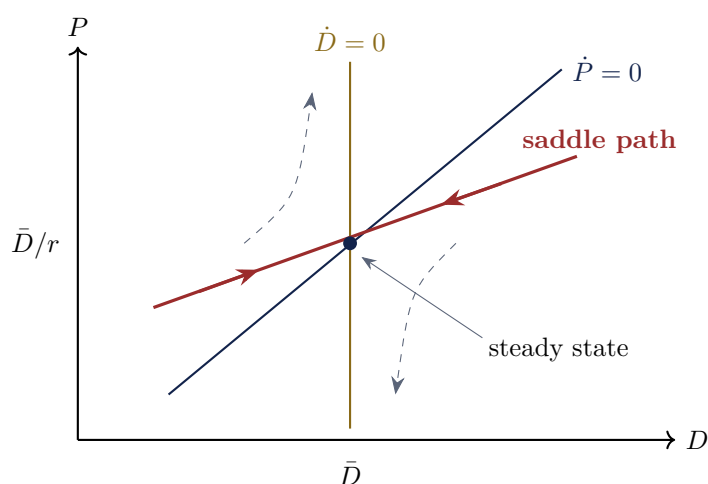


Figure 3: Phase diagram for (9). The dividend is predetermined and moves horizontally towards $\bar{D}$. The price jumps vertically onto the saddle path, which is the only trajectory that converges. Dashed paths are the divergent alternatives ruled out by the transversality condition (7).

Where this reappears

This is the same argument as the bubble discussion in §2. The unstable root and the bubble are the same object seen in continuous and discrete time; the transversality condition and the jump-variable selection are the same restriction. Every asset-pricing model in Movement II of the course uses this logic somewhere — and, as the next section shows, so does every optimal-control problem solved with a Hamiltonian: the control variable is the jump variable, chosen so that the system lands on the stable manifold.

4 The Hamiltonian and its costate variable

The consumption-and-portfolio lecture built the discrete-time method: a Lagrangian over the whole horizon, first-order conditions at every date, and a multiplier eliminated to produce an Euler equation. This section builds its continuous-time counterpart. It is the method behind Merton's continuous-time portfolio problem in Unit 1, Week 2, and every other continuous-time optimisation in the course.

Quick refresher: the discrete-time method, restated

There, a whole-horizon Lagrangian

$$\mathcal{L} = \mathbb{E} \left[\sum_t \beta^t r(x_t, u_t) + \sum_t \beta^{t+1} \psi_{t+1} (f(x_t, u_t, \varepsilon_{t+1}) - x_{t+1}) \right]$$

gave first-order conditions

$$\frac{\partial r}{\partial u_t} + \beta \mathbb{E}_t \left[\psi_{t+1} \frac{\partial f}{\partial u_t} \right] = 0, \quad \psi_t = \frac{\partial r}{\partial x_t} + \beta \mathbb{E}_t \left[\psi_{t+1} \frac{\partial f}{\partial x_t} \right],$$

whose second equation, with ψ eliminated between adjacent dates, is the Euler equation. Applied to consumption–saving with a constant return R , this reduced to

$$u'(c_t) = \beta \mathbb{E}_t [R u'(c_{t+1})] \quad (11)$$

— the consumption Euler equation. Everything below is the continuous-time limit of exactly this: the sum becomes an integral, the multiplier ψ_t becomes a continuously evolving costate ψ , and the second first-order condition above becomes the costate equation.

On method

Everything in this section is the continuous-time twin of the discrete-time method just restated. Wherever you see a costate equation, ask what the corresponding discrete-time multiplier-updating equation was; wherever you see the current-value Hamiltonian, ask what the corresponding whole-horizon Lagrangian was. The translation is exact, and keeping it in view is the fastest way to hold onto what follows.

4.1 A concrete problem: consumption from a productive asset

An agent holds wealth W_t invested in a technology with a concave return schedule $f(W)$ — $f' > 0$, $f'' < 0$ — and chooses a consumption path to solve

$$\max_{\{c_t\}} \int_0^\infty e^{-\rho t} u(c_t) dt \quad \text{subject to} \quad \dot{W} = f(W) - c, \quad W_0 \text{ given}, \quad (12)$$

with $u(c) = c^{1-\gamma}/(1-\gamma)$ and $\rho > 0$ the rate of time preference.

In plain English

“Concave return schedule” means: the more wealth already invested, the less extra output each additional unit produces — diminishing returns, exactly as in the production functions from your microeconomics course. This is what will produce a genuine, two-dimensional saddle below, in the same way the mean-reverting dividend did in §3: without diminishing returns, there would be nothing to pull the system back toward a steady state.

This is the continuous-time consumption–saving problem, with a diminishing-returns investment opportunity in place of the constant-return riskless bond of the discrete-time method.

4.2 The current-value Hamiltonian

Quick refresher

From a sum to an integral. The discrete-time Lagrangian restated above summed a payoff plus a multiplier-weighted constraint over every date, $t = 0, 1, \dots, T$. In continuous time, “every date” becomes a continuum, the sum becomes an integral, and the constraint — attached with a multiplier at every instant — becomes the state equation $\dot{x} = f(x, u)$. The *current-value Hamiltonian* defined below is exactly the integrand of that continuous-time Lagrangian, isolated at a single instant: it plays precisely the role that the term inside the summation played in the discrete-time Lagrangian.

Definition 4.1 (Current-value Hamiltonian). For a problem of the form $\max_u \int_0^\infty e^{-\rho t} r(x, u) dt$ subject to $\dot{x} = f(x, u)$, the current-value Hamiltonian is

$$H(x, u, \psi) = r(x, u) + \psi f(x, u), \quad (13)$$

where ψ is the **costate variable** — the continuous-time multiplier attached to the state equation, playing exactly the role ψ_t played in the discrete-time Lagrangian.

In plain English

H adds up two things happening at the same instant: the direct payoff $r(x, u)$ from today’s choice, plus the *value* of how that choice is changing the state, $f(x, u)$, priced at the shadow price ψ . Maximising H over u therefore balances “how good is this choice right now” against “how much is this choice helping or hurting my future wealth, measured in today’s units.”

Theorem 4.2 (Pontryagin’s conditions, current-value form). *An interior optimum of the problem above satisfies, at every t ,*

$$\frac{\partial H}{\partial u} = 0, \quad (14)$$

$$\dot{\psi} = \rho \psi - \frac{\partial H}{\partial x}, \quad (15)$$

$$\dot{x} = \frac{\partial H}{\partial \psi} = f(x, u), \quad (16)$$

together with the transversality condition $\lim_{t \rightarrow \infty} e^{-\rho t} \psi_t x_t = 0$.

In plain English

Three conditions, three jobs. (14) picks the best control given today's shadow price: adjust u until its direct marginal payoff exactly balances its marginal effect on tomorrow's wealth. (15) says how the shadow price itself moves: it grows at the discount rate ρ minus whatever direct marginal benefit the state itself provides. (16) simply restates that the state obeys its own law of motion — included here only so the system is complete. The transversality condition rules out the shadow-price-weighted state exploding as $t \rightarrow \infty$, exactly as (7) ruled out an exploding bubble.

These three conditions are the exact continuous-time analogues of the two discrete-time first-order conditions restated above: (14) is the control's first-order condition, (15) is the law of motion for the multiplier (the costate equation), and (16) simply restates the constraint. No value function is defined, and (15) is not an envelope-theorem argument on one — it is the continuous-time limit of the discrete-time multiplier-updating equation, nothing more.

Common confusion

Students who have seen the Hamilton–Jacobi–Bellman (HJB) equation elsewhere sometimes expect ψ to secretly be $V'(x)$ for some value function V . Under enough regularity, the two approaches *can* be shown to agree — Pontryagin's maximum principle and dynamic programming are equivalent theories in continuous time, just as their discrete-time counterparts are. But nothing in the derivation above defines, or requires, V . The three conditions (14)–(16) are self-contained: they follow from treating ψ as a Lagrange multiplier on a continuum of constraints, exactly as in the discrete-time Euler-equation method, and never invoke a function evaluated away from the realised path.

4.3 Applying it to the consumption problem

For (12), $H(W, c, \psi) = u(c) + \psi(f(W) - c)$. Condition (14) gives

$$u'(c) = \psi, \quad (17)$$

which reads exactly like the discrete-time result $\psi_t = u'(c_t)$: the shadow price of wealth equals the marginal utility of consumption, because at an optimum the agent must be indifferent between consuming the marginal unit and investing it. Condition (15) gives the costate equation

$$\dot{\psi} = \rho\psi - \psi f'(W) = \psi(\rho - f'(W)). \quad (18)$$

On method

Compare (18) with the discrete-time consumption Euler equation (11) above. There, $u'(c_t)/u'(c_{t+1})$ measured how fast the shadow price needed to fall between two dates for the agent to be willing to postpone consumption, given the return R . Here, $\dot{\psi}/\psi = \rho - f'(W)$ measures the same thing *instantaneously*: the costate falls over time exactly when the return $f'(W)$ exceeds the discount rate ρ , because a high return makes waiting attractive enough that the marginal unit of wealth is worth less in present terms with every instant it is not consumed.

In plain English

If the return on wealth $f'(W)$ is high relative to how impatient the agent is (ρ), the shadow price of wealth is falling over time — meaning wealth is becoming relatively more abundant, in utility terms, as it compounds faster than the agent's impatience discounts it. If $f'(W)$ is low relative to ρ , the opposite: the shadow price rises, because compounding is not keeping pace with impatience.

4.4 Eliminating the costate: a system in the state and the control**Quick refresher**

Why eliminate ψ at all? The pair $(\dot{W}, \dot{\psi})$ from (15)–(16) is already a two-dimensional system, and you could in principle draw a phase diagram directly in (W, ψ) -space. But ψ is an artificial, unobservable object — nobody quotes a market price for the marginal utility of wealth. Converting the system into (W, c) -space, where c is an observable, economically meaningful quantity, is purely a matter of convenience, made possible because (17) gives a one-to-one link between ψ and c at every instant.

Differentiate (17) with respect to time: $u''(c)\dot{c} = \dot{\psi}$. Substituting (18) and using $u'(c) = \psi$,

$$u''(c)\dot{c} = u'(c)(\rho - f'(W)) \implies \dot{c} = \frac{u'(c)}{u''(c)}(\rho - f'(W)).$$

Quick refresher

The relative risk aversion coefficient. For CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$, $u'(c) = c^{-\gamma}$ and $u''(c) = -\gamma c^{-\gamma-1}$, so $-cu''(c)/u'(c) = -c \cdot (-\gamma c^{-\gamma-1})/c^{-\gamma} = \gamma$: this particular combination of derivatives, weighted by c , is a constant, equal to γ , for every consumption level — which is exactly the “constant relative risk aversion” the name promises, and exactly the same property that made the myopia result of the consumption-and-portfolio lecture go through. It follows immediately that $u'(c)/u''(c) = -c/\gamma$.

For CRRA utility, $-cu''(c)/u'(c) = \gamma$, so $u'(c)/u''(c) = -c/\gamma$, and the costate has been eliminated entirely in favour of a system in (W, c) :

$$\dot{W} = f(W) - c, \quad \dot{c} = \frac{c}{\gamma}(f'(W) - \rho). \quad (19)$$

This is precisely the form of (8): a two-dimensional autonomous dynamical system, ready for exactly the trace–determinant analysis built in §3.

In plain English

Read (19) as two rules. Wealth grows by whatever is produced, $f(W)$, minus whatever is consumed, c — an accounting identity. Consumption grows (or shrinks) at a rate proportional to the gap between the return on wealth $f'(W)$ and the discount rate ρ : if wealth is earning more than the agent's impatience requires, she lets consumption grow over time rather than consuming it all now; if it is earning less, she front-loads consumption.

4.5 Steady state and stability

The steady state solves $\dot{c} = 0$ and $\dot{W} = 0$:

$$f'(W^*) = \rho \quad (\text{the modified golden rule}), \quad c^* = f(W^*). \quad (20)$$

In plain English

At the steady state, the return on wealth exactly equals the discount rate — if it were higher, the agent would still want to be shifting consumption into the future (so c would still be changing, contradicting $\dot{c} = 0$); if lower, she would want to be front-loading it. And wealth is constant only if consumption exactly matches what is produced, $c^* = f(W^*)$: no more, no less.

Proposition 4.3 (The consumption–wealth system is a saddle). *The Jacobian of (19) at the steady state (20) is*

$$\mathbf{J} = \begin{pmatrix} f'(W^*) & -1 \\ \frac{c^*}{\gamma} f''(W^*) & 0 \end{pmatrix}, \quad \Delta = \det \mathbf{J} = \frac{c^*}{\gamma} f''(W^*) < 0,$$

so the system has a saddle point at (W^*, c^*) , with a one-dimensional stable manifold.

Proof. Step 1: differentiate the right-hand sides of (19). $\partial \dot{W} / \partial W = f'(W)$ and $\partial \dot{W} / \partial c = -1$ directly. For the second row, $\partial \dot{c} / \partial c = (f'(W) - \rho) / \gamma$ and $\partial \dot{c} / \partial W = (c / \gamma) f''(W)$.

Step 2: evaluate at the steady state. $\partial \dot{c} / \partial c$ vanishes at $W = W^*$ by (20) (since $f'(W^*) = \rho$), giving the zero in the bottom right of $\mathbf{J}$.

Step 3: compute the determinant. Δ is the product of the diagonal minus the product of the off-diagonal: $\Delta = f'(W^*) \cdot 0 - (-1) \cdot (c^* / \gamma) f''(W^*) = (c^* / \gamma) f''(W^*)$, negative because $f'' < 0$, $c^* > 0$ and $\gamma > 0$.

Step 4: read off the classification. By the trace–determinant table of §3, $\Delta < 0$ is exactly the saddle condition, regardless of the trace. $\square$

Exactly as with the price in §3, consumption c is the jump variable here — it can move instantly, while wealth W is predetermined by past accumulation and moves only continuously. The economy selects the value of c_0 that places (W_0, c_0) exactly on the stable manifold; any other choice sends the system either to $c = 0$ with wealth still positive (sub-optimally hoarding) or to $W = 0$ in finite time (starvation), both ruled out by the transversality condition.

In plain English

If the economy started with too little consumption relative to the saddle path, wealth would keep accumulating forever with consumption never catching up — the agent would be hoarding, working hard and never enjoying the proceeds, which cannot be optimal. If it started with too much consumption, wealth would be run down to zero in finite time, after which nothing more can be consumed — also clearly not optimal, since a slightly lower initial consumption would have avoided ever running out. The saddle path is the unique compromise between these two failures.

Example 4.4 (A numerical steady state). Let $f(W) = AW^\alpha$ with $A = 0.5$, $\alpha = 0.35$, $\rho = 0.04$ and $\gamma = 2$. The modified golden rule gives

$$f'(W^*) = \alpha A (W^*)^{\alpha-1} = \rho \implies W^* = \left(\frac{\alpha A}{\rho} \right)^{1/(1-\alpha)} = \left(\frac{0.35 \times 0.5}{0.04} \right)^{1/0.65} \approx 9.69,$$

so $c^* = f(W^*) = 0.5 \times 9.69^{0.35} \approx 1.107$. At this steady state, $f''(W^*) = \alpha(\alpha - 1)A(W^*)^{\alpha-2} \approx -0.00268$, giving $\Delta = (c^*/\gamma)f''(W^*) \approx (1.107/2) \times (-0.00268) \approx -0.00149 < 0$, confirming the saddle.

Where this reappears

This is exactly the method Unit 1, Week 2 uses for Merton's continuous-time portfolio problem: a current-value Hamiltonian in wealth, a costate equation, and a two-equation dynamical system whose stability is read off the trace-determinant plane. There, the state is wealth (or, in the version with a mean-reverting opportunity set, the state expands to include a mean-reverting variable driving the expected return, exactly as in the consumption-and-portfolio lecture's discussion of a moving opportunity set), the costate is the shadow price of wealth, and the portfolio share decomposes into a myopic term — present even when the costate system collapses to the degenerate, one-dimensional case of Samuelson's myopia theorem — and an intertemporal hedging term, read directly off the costate dynamics whenever it does not.

5 Stochastic processes

The last piece of apparatus. Everything here is used from Week 2 onwards.

5.1 The random walk

Definition 5.1. $\{x_t\}$ is a random walk with drift μ if $x_{t+1} = \mu + x_t + \varepsilon_{t+1}$ with ε_t i.i.d. mean zero and variance σ^2 .

Iterating, $x_{t+h} = x_t + \mu h + \sum_{j=1}^h \varepsilon_{t+j}$, so

$$\mathbb{E}_t[x_{t+h}] = x_t + \mu h, \quad \text{Var}_t(x_{t+h}) = h\sigma^2.$$

In plain English

The expected level moves in a straight line, μ per period, exactly as you would guess. The *variance* of where you end up grows *linearly* in the horizon — not because uncertainty compounds multiplicatively, but because the shocks are independent, and variances of independent things simply add. The standard deviation therefore grows with the *square root* of the horizon, which is why doubling your forecast horizon does not double your uncertainty — it multiplies it by only $\sqrt{2} \approx 1.41$.

Variance grows *linearly* in the horizon — the square-root-of-time rule. This is the null hypothesis the variance ratio tests in Unit 2 (Weeks 5–6), and the benchmark against which the horizon effects of Unit 1 (Week 2) are measured.

5.2 The AR(1) process and mean reversion

Definition 5.2. $x_{t+1} = \mu + \phi x_t + \varepsilon_{t+1}$ is a first-order autoregression. It is covariance stationary if $|\phi| < 1$.

In plain English

This is the random walk's better-behaved cousin: instead of drifting forever, x_t is pulled back toward a long-run mean $\mu/(1-\phi)$ at a rate governed by ϕ . It is literally equation (1) from §1, with $a = \phi$ and $b = \mu$, but with a random shock added each period instead of the deterministic path being exact.

Proposition 5.3. *If $|\phi| < 1$ then*

$$\mathbb{E}[x] = \frac{\mu}{1-\phi}, \quad \text{Var}(x) = \frac{\sigma^2}{1-\phi^2}, \quad \text{Corr}(x_t, x_{t-k}) = \phi^k.$$

The half-life of a shock — the horizon h at which its effect has decayed by half — is $h = \ln(0.5)/\ln \phi$.

Example 5.4. For monthly data with $\phi = 0.9$, the half-life is $\ln(0.5)/\ln(0.9) = 6.58$ months and the unconditional standard deviation is 2.294σ . For $\phi = 0.98$ the half-life is 34.3 months and the unconditional standard deviation 5.025σ . A small change in the persistence parameter produces a very large change in the character of the series — which is why estimates of ϕ close to one are both empirically common and statistically treacherous.

Where this reappears

The AR(1) is the workhorse of the whole course. In Unit 1 (Week 2) it drives the expected return and generates the horizon effects that break Samuelson's myopia theorem, covered in the consumption-and-portfolio lecture. In Unit 4 (Week 9) it is the short interest rate in the Vasicek model, where ϕ becomes the speed of mean reversion. In Unit 2 (Week 4) it is the analogue of the conditional variance in the GARCH recursion.

5.3 Martingales and the law of iterated expectations

Definition 5.5. $\{x_t\}$ is a martingale with respect to the filtration $\{\mathcal{I}_t\}$ if $\mathbb{E}[|x_t|] < \infty$ and $\mathbb{E}[x_{t+1} | \mathcal{I}_t] = x_t$ for all t .

The single most useful property in this course is the tower property: for $\mathcal{I}_t \subseteq \mathcal{I}_{t+1}$,

$$\mathbb{E}[\mathbb{E}[X | \mathcal{I}_{t+1}] | \mathcal{I}_t] = \mathbb{E}[X | \mathcal{I}_t]. \quad (21)$$

In plain English

The tower property just says: averaging a forecast that already used more information, using only less information, gives you back the forecast you would have made with the less information in the first place. There is no way to improve on your best current guess by first pretending to know more, then averaging that pretence back down.

Common confusion

Note what a martingale restricts and what it does not. It is a condition on the conditional *mean* and on nothing else. It permits the variance to change over time, it permits dependence in higher moments, and it therefore permits volatility clustering. Conflating “martingale” with “independent and identically distributed” is the most common error in this material, and it produces bad arguments in both directions — see Unit 2 (Weeks 5–6).

5.4 Brownian motion and Itô's lemma

For the continuous-time work in Unit 1 we need one result, stated without proof.

Definition 5.6. A standard Brownian motion $\{W_t\}$ has $W_0 = 0$, independent increments, and $W_{t+h} - W_t \sim N(0, h)$, with continuous sample paths.

Theorem 5.7 (Itô's lemma, one dimension). *If $dX_t = a(X_t, t) dt + b(X_t, t) dW_t$ and f is twice continuously differentiable, then*

$$df(X_t, t) = \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx} \right) dt + b f_x dW_t. \quad (22)$$

Quick refresher

Why ordinary calculus is not enough here. For an ordinary (non-random) function, the chain rule has no second-derivative term at first order: $df(X_t) = f_x dX_t$ is enough, because $(dX_t)^2$ is negligible compared to dX_t itself as the time step shrinks. For Brownian motion this stops being true: over a small interval of length dt , a discretised increment has variance dt , and so $(dW_t)^2$ has the same *order of magnitude* as dt itself, not something smaller. Informally, $(dW)^2 = dt$. That extra, non-negligible term is exactly what produces the second-derivative correction in (22).

The term $\frac{1}{2}b^2 f_{xx}$ has no counterpart in ordinary calculus.

Example 5.8 (Geometric Brownian motion). Let $dS = \mu S dt + \sigma S dW$ and take $f = \ln S$. Then $f_x = 1/S$ and $f_{xx} = -1/S^2$, so (22) gives

$$d \ln S = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW.$$

The drift of the log price is *lower* than the drift of the price by $\sigma^2/2$.

In plain English

This is not a technicality. It is why the expected growth rate of an investment and its median (typical) growth rate differ, and why volatility is costly to a long-horizon investor even when it is symmetric: an asset that goes up 20% then down 20% has not returned to its starting value — it has *lost* $1 - 1.2 \times 0.8 = 4\%$ — and Itô's correction term is exactly what is tracking that asymmetry.

6 Nonlinearity previewed

Everything above is linear (or linearised near a steady state), and linear systems have a limited behavioural repertoire: converge, diverge, or oscillate at a fixed frequency. Nonlinear systems can do more, and three possibilities matter later in the course.

- **Multiple equilibria.** $f(x^*) = x^*$ may have several solutions, some stable and some not. Which one is reached depends on the initial condition — history matters in a way it cannot in a linear model.
- **Limit cycles.** A nonlinear planar system can possess an isolated closed orbit which nearby trajectories approach. The system then oscillates indefinitely with an amplitude determined by the system itself rather than by initial conditions. This is used in Unit 5 for the Goodwin–Keen model.

- **Bifurcation and chaos.** As a parameter varies, the qualitative character of the dynamics can change discontinuously. In the *Hopf bifurcation*, a stable focus loses stability and a limit cycle is born as a pair of complex eigenvalues crosses the imaginary axis. Under further parameter variation, period-doubling can lead to deterministic chaos: bounded, aperiodic paths with sensitive dependence on initial conditions.

Where this reappears

The Hopf bifurcation appears twice more: in Unit 2 (Weeks 5–6), where an increase in the intensity with which agents switch between forecasting rules destabilises the price and produces chaotic dynamics in the Brock–Hommes model; and in Unit 5 (Week 11), where adding debt to the Goodwin system generates limit cycles and, past a threshold, breakdown. Both are deterministic systems producing irregular paths — a reminder that irregularity in data does not, by itself, imply an underlying random process.

7 Exercises

Exercise 1. Solve $x_{t+1} = 0.8x_t + 3$ with $x_0 = 20$. Find the steady state, write down the closed-form path, and compute the half-life of the deviation. Confirm your answer against a simulation.

Exercise 2. For the second-order equation $x_{t+2} - 1.2x_{t+1} + 0.85x_t = 0.65$, find the characteristic roots, classify the dynamics, compute the period of oscillation, and determine whether the steady state is stable. Verify the Schur conditions.

Exercise 3. Take the pricing equation (5) with $r = 0.05$ and $D_t = 2$ for all t .

- Verify $F = 40$.
- An investor buys at $P_0 = 41$ on a path with a bubble. What is the price at $t = 30$? What fraction of the price is bubble?
- Explain why no arbitrage argument available *within* the model rules this path out. What extra assumption is required, and what is its status?

Exercise 4. For the system (9) with $\bar{D} = 3$, $r = 0.04$ and $g = 0.25$: compute the steady state, verify that the Jacobian has a negative determinant, derive the saddle path, and evaluate the price when $D = 2$. Confirm your answer by computing the present value directly.

Exercise 5. In the model of Proposition 3.3, suppose $g \rightarrow 0$ (dividends become a random walk rather than mean-reverting). What happens to the slope of the saddle path, and why? Interpret economically.

Exercise 6. For the Hamiltonian system (19) with $f(W) = AW^\alpha$, $A = 0.4$, $\alpha = 0.4$, $\rho = 0.05$ and $\gamma = 3$: compute the steady state (W^*, c^*) , verify $\Delta = \det \mathbf{J} < 0$, and sketch the phase diagram, marking the saddle path and one divergent path. What does the transversality condition rule out economically at each end of the divergent path?

Exercise 7. An AR(1) has $\phi = 0.95$ at monthly frequency. Compute the half-life in months and the unconditional variance in terms of σ^2 . Now suppose you have 120 observations. Comment, informally, on how precisely you expect to be able to estimate ϕ , and why this matters for the portfolio conclusions of Unit 1.

Exercise 8. Use Itô's lemma to find the process followed by $f(S) = S^2$ when $dS = \mu S dt + \sigma S dW$. Where does the correction term come from, and what would ordinary calculus have given?

Exercise 9. Give an example of a process that is a martingale but not a random walk with i.i.d. increments. Explain in two sentences why the distinction matters for testing market efficiency.

Reading

- Shone, R. (2002). *Economic Dynamics: Phase Diagrams and Their Economic Application*, 2nd edn, Ch. 2–4.
- Merton, R. C. (1969). Lifetime portfolio selection under uncertainty: the continuous-time case. *Review of Economics and Statistics* 51(3) — Section 1 only, for the framing of the problem.
- Medio, A. & Lines, M. (2001). *Nonlinear Dynamics: A Primer*, Ch. 1 and 3.
- Léonard, D. & Van Long, N. (1992). *Optimal Control Theory and Static Optimization in Economics*, Ch. 1–3, 6 — for the Hamiltonian and Pontryagin's maximum principle in the form used here.

This is Lecture 4 of 4 covering the Foundations material. Lecture 1 covers why time changes the problem and the map of the course. Lecture 2 covers why repeating the static solution is not the dynamic solution, and the Euler-equation method. Lecture 3 covers optimum versus trajectory and the question of stability.

Numerical values in the worked examples have been computed and cross-checked; they are constructed for teaching and are not drawn from market data. Figures are schematic.